

Global Earthquake & Tsunami Risk Assessment

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Projeto Final - Ferramentas Computacional de Modelagem

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- 1 Conjunto de dados
- 2 Descrição das variáveis
- 3 Análises
- 4 Conclusão

Conjunto de dados



Global Earthquake & Tsunami Risk Assessment

Ahmed Mohamed Zaki

- Conjunto de dados de avaliação de risco de terremoto e tsunami
- Informações geoespaciais, características sísmicas e indicadores de potencial de tsunami
- 782 terremotos registrados globalmente entre os anos de 2001 e 2022



Variáveis

Tabela 1: Descrição das variáveis do conjunto de dados

Variável	Tipo	Descrição	Intervalo
magnitude	<i>double</i>	Magnitude do terremoto (escala Richter)	6.5 – 9.1
cdi	<i>integer</i>	Medida do impacto populacional (Community Decimal Intensity)	0 – 9
mmi	<i>integer</i>	Indicador de danos estruturais (Modified Mercalli Intensity)	1 – 9
sig	<i>integer</i>	Score de significância do evento	650 – 2910
nst	<i>integer</i>	Número de estações de monitoramento sísmico	0 – 934
dmin	<i>double</i>	Distância do evento até a estação sísmica mais próxima (graus)	0.0 – 17.7
gap	<i>double</i>	Lacuna azimutal entre estações (graus)	0.0 – 239.0
depth	<i>double</i>	Profundidade do foco do terremoto (km)	2.7 – 670.8
latitude	<i>double</i>	Latitude do epicentro (WGS84)	-61.85° – $+71.63^{\circ}$
longitude	<i>double</i>	Longitude do epicentro (WGS84)	-179.97° – $+179.66^{\circ}$
Year	<i>integer</i>	Ano de ocorrência	2001 – 2022
Month	<i>integer</i>	Mês de ocorrência	1 – 12
tsunami	<i>integer</i>	Potencial de tsunami	0 = não; 1 = sim

Pacotes e leitura do arquivo

```
library(tidyverse)
library(maps)

dados <- read.csv("earthquake_data_tsunami.csv")

head(dados)
```

	magnitude	cdi	mmi	sig	nst	dmin	gap	depth	latitude	longitude	Year	Month	tsunami
1	7.0	8	7	768	117	0.509	17	14.000	-9.7963	159.596	2022	11	1
2	6.9	4	4	735	99	2.229	34	25.000	-4.9559	100.738	2022	11	0
3	7.0	3	3	755	147	3.125	18	579.000	-20.0508	-178.346	2022	11	1
4	7.3	5	5	833	149	1.865	21	37.000	-19.2918	-172.129	2022	11	1
5	6.6	0	2	670	131	4.998	27	624.464	-25.5948	178.278	2022	11	1
6	7.0	4	3	755	142	4.578	26	660.000	-26.0442	178.381	2022	11	1

Pacotes e leitura do arquivo

```
glimpse(dados)
```

```

Rows: 782
Columns: 13
$ magnitude <dbl> 7.0, 6.9, 7.0, 7.3, 6.6, 7.0, 6.8, 6.7, 6.8, 7.6, 6.9, 6.5, 7.0, 7.6,...
$ cdi       <int> 8, 4, 3, 5, 0, 4, 1, 7, 8, 9, 9, 7, 7, 8, 9, 7, 9, 3, 7, 2, 6, 6, 6, 6,...
$ mmi       <int> 7, 4, 3, 5, 2, 3, 3, 6, 7, 8, 9, 7, 5, 8, 8, 6, 8, 2, 5, 5, 5, 5, 4, 4,...
$ sig       <int> 768, 735, 755, 833, 670, 755, 711, 797, 1179, 1799, 887, 756, 761,...
$ nst       <int> 117, 99, 147, 149, 131, 142, 136, 145, 175, 271, 215, 178, 192, 272,...
$ dmin      <dbl> 0.509, 2.229, 3.125, 1.865, 4.998, 4.578, 4.678, 1.151, 2.137,...
$ gap       <dbl> 17, 34, 18, 21, 27, 26, 22, 37, 92, 69, 34, 54, 45, 12, 34, 34, 22, 31,...
$ depth     <dbl> 14.000, 25.000, 579.000, 37.000, 624.464, 660.000, 630.379, 20.000,...
$ latitude  <dbl> -9.7963, -4.9559, -20.0508, -19.2918, -25.5948, -26.0442, -25.9678,...
$ longitude <dbl> 159.5960, 100.7380, -178.3460, -172.1290, 178.2780, 178.3810,...
$ Year      <int> 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022,...
$ Month     <int> 11, 11, 11, 11, 11, 11, 10, 9, 9, 9, 9, 9, 9, 9, 8, 7, 6, 5, 5, 5,...
$ tsunami  <int> 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 1, 1, 1, 1, 1,...

```

Análise descritiva

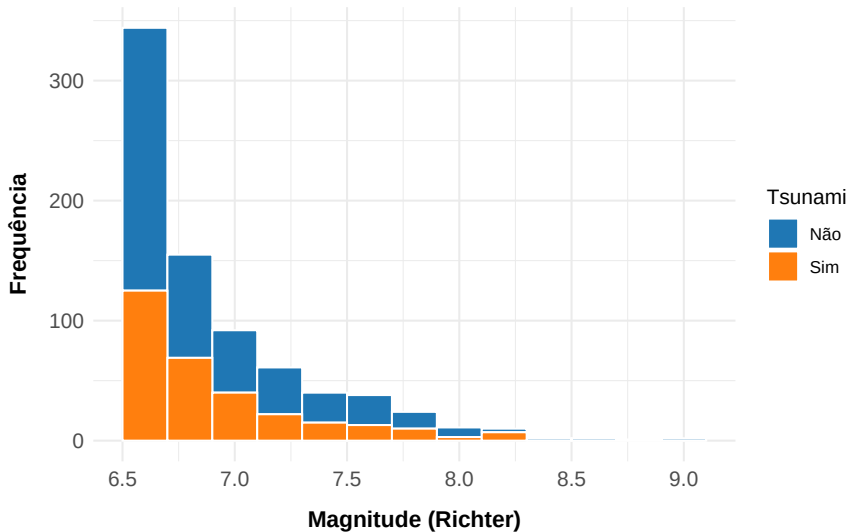
```
dados %>%
  group_by(tsunami) %>%
  summarise(
    n = n(),
    media_magnitude = mean(magnitude, na.rm = TRUE),
    mediana_magnitude = median(magnitude, na.rm = TRUE),
    sd_magnitude = sd(magnitude, na.rm = TRUE),
    media_depth = mean(depth, na.rm = TRUE),
    mediana_depth = median(depth, na.rm = TRUE),
    sd_depth = sd(depth, na.rm = TRUE)
  )
```

tsunami	n	magnitude			profundidade		
		média	mediana	sd	média	mediana	sd
0	478	6.942803	6.8	0.4595409	69.66736	26.0000	127.5012
1	304	6.938487	6.8	0.4232504	85.65680	26.9715	151.0803

Distribuição das magnitudes

```
ggplot(dados, aes(x = magnitude, fill = tsunami)) +
  geom_histogram(binwidth = 0.2, color = "white") +
  labs(
    title = "Distribuicao das magnitudes dos terremotos (2001 - 2022)",
    x = "Magnitude (Richter)",
    y = "Frequencia"
  ) +
  scale_fill_manual(values = c("#1f77b4", "#ff7f0e"),
    name = "Tsunami", labels = c("0" = "Nao", "1" = "Sim")) +
  theme_minimal()+
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 12, face = "bold"),
    axis.title.y = element_text(size = 12, face = "bold"),
    axis.text = element_text(size = 11)
  )
```

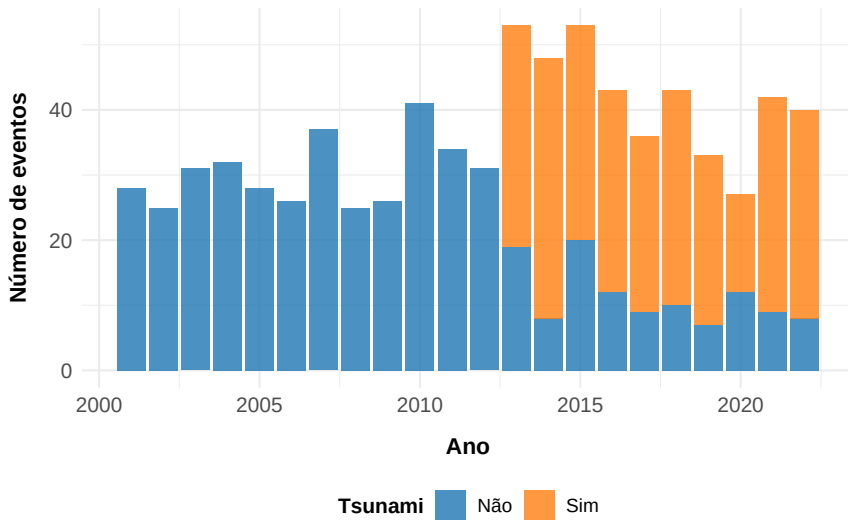
Distribuição das magnitudes dos terremotos (2001–2022)



Número de eventos por ano

```
dados %>%
  count(Year, tsunami) %>%
  ggplot(aes(x = Year, y = n, fill = tsunami)) +
  geom_col(position = position_stack(reverse = TRUE), alpha = 0.8) +
  scale_fill_manual(values = c("#1f77b4", "#ff7f0e"), name = "Tsunami",
                    labels = c("0" = "Nao", "1" = "Sim")) +
  labs(title = "Numero de Eventos Sismicos por Ano (2001 - 2022)",
       x = "Ano", y = "Numero de eventos")+
  theme_minimal()+
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 12, face = "bold"),
    axis.title.y = element_text(size = 12, face = "bold"),
    axis.text = element_text(size = 11),
    legend.position = "bottom",
    legend.text = element_text(size = 10),
    legend.title = element_text(size = 11, face = "bold")
  )
```

Número de Eventos Sísmicos por Ano (2001 – 2022)



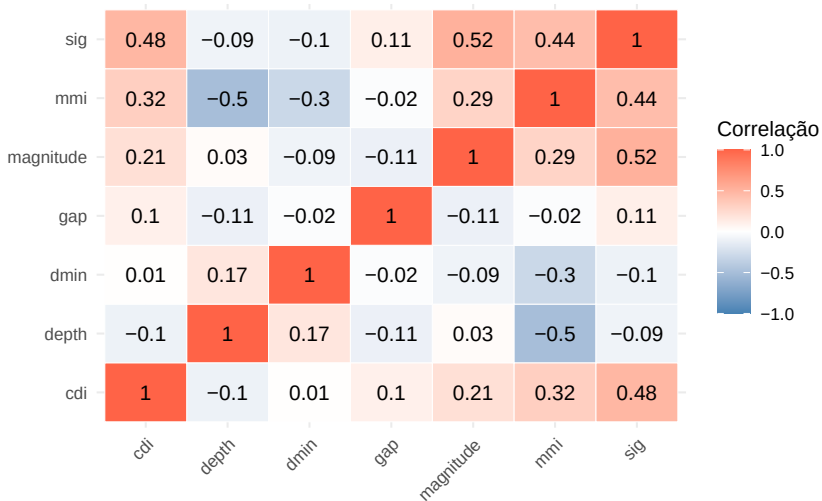
Correlação entre variáveis

```
vars_corr <- dados %>%  
  select(magnitude, depth, cdi, mmi, sig, dmin, gap)  
  
# Matriz de correlacao  
corr_mat <- cor(vars_corr, use = "pairwise.complete.obs") %>%  
  round(2)  
  
# Converter pra formato longo  
corr_long <- corr_mat %>%  
  as.data.frame() %>%  
  rownames_to_column("var1") %>%  
  pivot_longer(-var1, names_to = "var2", values_to = "cor")
```

Correlação entre variáveis

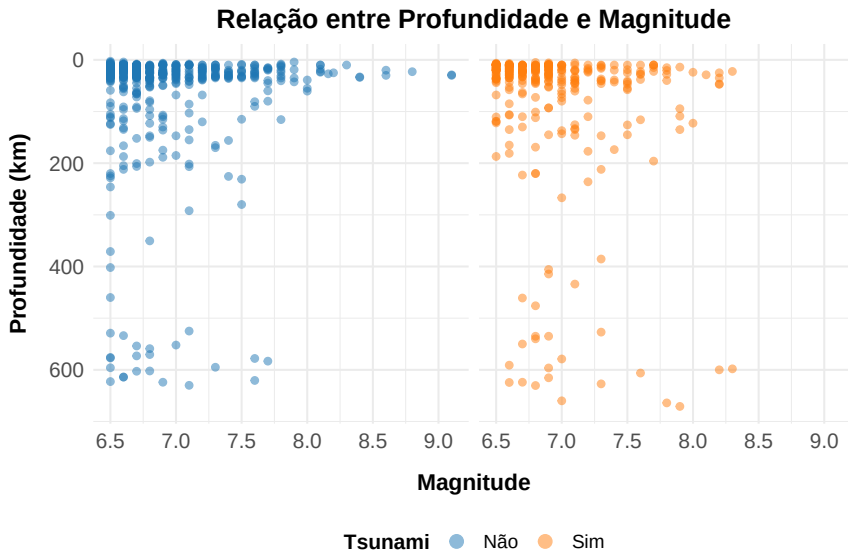
```
ggplot(corr_long, aes(var1, var2, fill = cor)) +  
  geom_tile(color = "white") +  
  geom_text(aes(label = cor), size = 4) +  
  scale_fill_gradient2(  
    low = "steelblue", mid = "white", high = "tomato",  
    midpoint = 0, limits = c(-1, 1)) +  
  labs(  
    title = "Correlacao entre Variaveis Sismicas", x = "", y = "",  
    fill = "Correlacao") +  
  theme_minimal() +  
  theme(  
    axis.text.x = element_text(angle = 45, hjust = 1, vjust = 1),  
    plot.title = element_text(hjust = 0.5, size = 14, face = "bold")  
  )
```

Correlação entre Variáveis Sísmicas



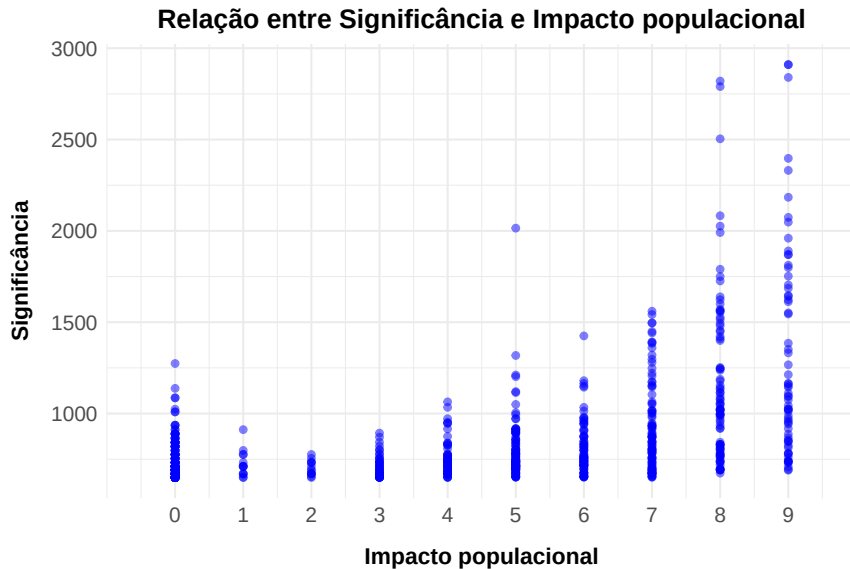
Profundidade vs. Magnitude

```
dados %>%
  group_by(tsunami) %>%
  ggplot(aes(x = magnitude, y = depth, color = tsunami)) +
  geom_point(alpha = 0.5) +
  scale_color_manual(values = c("#1f77b4", "#ff7f0e"), name = "Tsunami", labels
    = c("0" = "Nao", "1" = "Sim")) +
  labs(title = "Relacao entre Profundidade e Magnitude",
    y = "Profundidade (km)", x = "Magnitude")+
  scale_y_reverse() +
  guides(color = guide_legend(override.aes = list(size = 3)))+
  facet_wrap(~tsunami)+
  theme_minimal()+
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 12, face = "bold"),
    axis.title.y = element_text(size = 12, face = "bold"),
    axis.text = element_text(size = 11),
    legend.position = "bottom",
    legend.text = element_text(size = 10),
    legend.title = element_text(size = 11, face = "bold")
  )
```



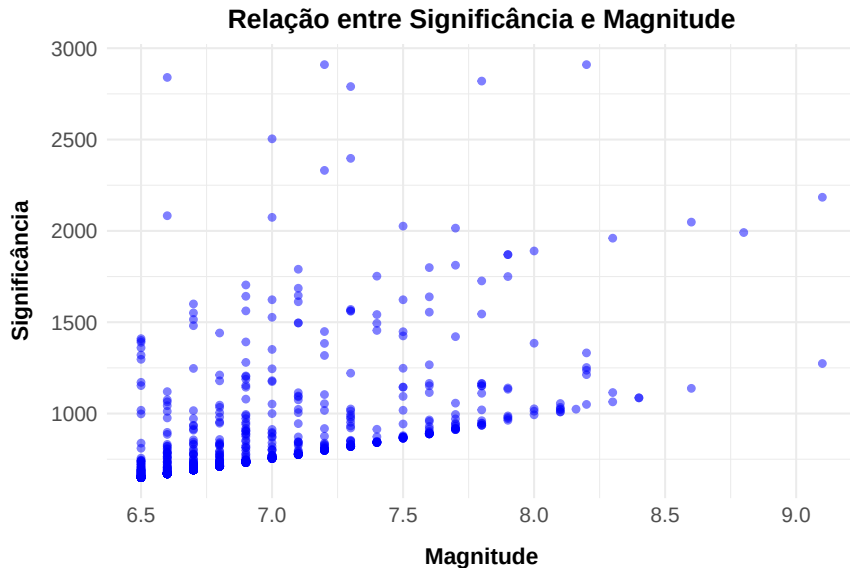
Significância vs. Impacto populacional

```
dados %>%
  ggplot(aes(y = sig, x = cdi)) +
  geom_point(alpha = 0.5, color = "blue") +
  scale_x_continuous(limits = c(-0.5, 9.5), breaks = 0:9) +
  labs(title = "Relacao entre Significancia e Impacto populacional",
       x = "Impacto populacional", y = "Significancia")+
  theme_minimal()+
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 12, face = "bold"),
    axis.title.y = element_text(size = 12, face = "bold"),
    axis.text = element_text(size = 11)
  )
```



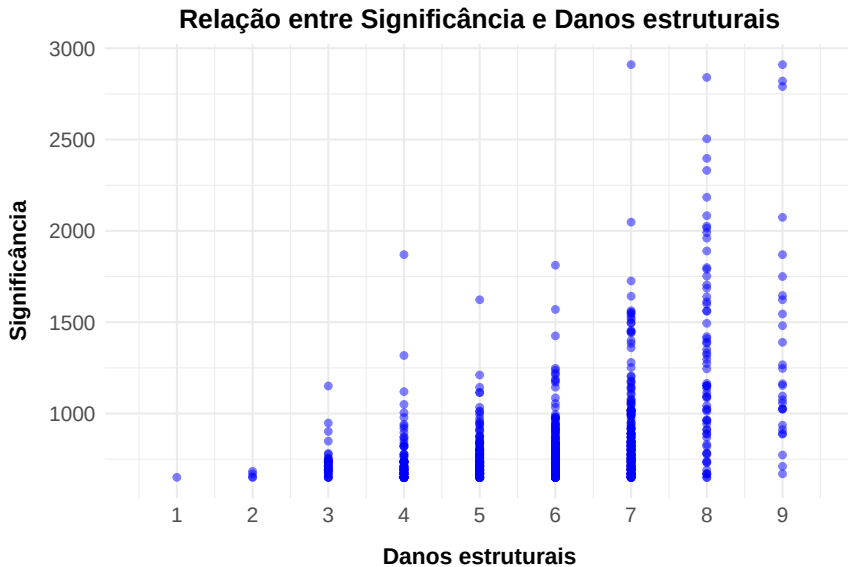
Significância vs. Magnitude

```
dados %>%  
  ggplot(aes(y = sig, x = magnitude)) +  
  geom_point(alpha = 0.5, color = "blue") +  
  labs(title = "Relacao entre Significancia e Magnitude",  
        x = "Magnitude", y = "Significancia")+  
  theme_minimal()+  
  theme(  
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),  
    axis.title.x = element_text(size = 12, face = "bold"),  
    axis.title.y = element_text(size = 12, face = "bold"),  
    axis.text = element_text(size = 11)  
  )
```



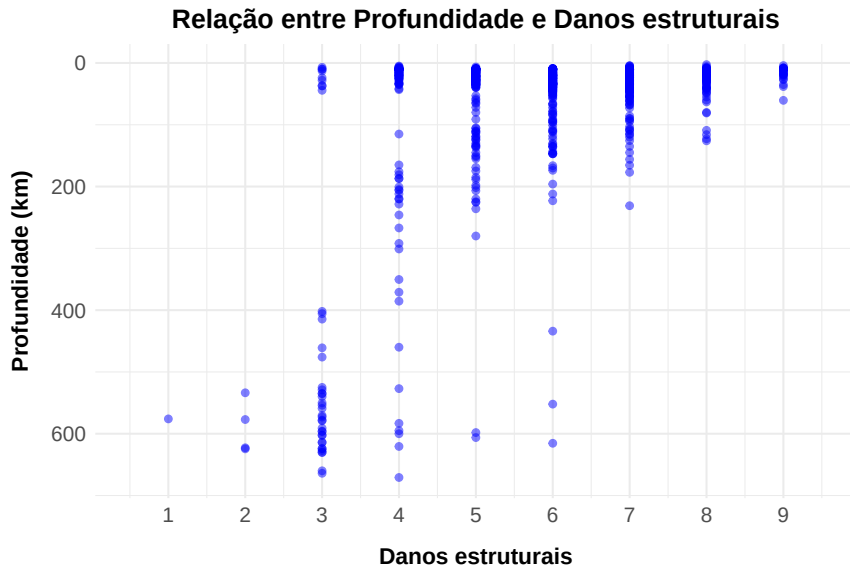
Significância vs. Danos estruturais

```
dados %>%
  ggplot(aes(y = sig, x = mmi)) +
  geom_point(alpha = 0.5, color = "blue") +
  scale_x_continuous(limits = c(0.5, 9.5), breaks = 1:9) +
  labs(title = "Relacao entre Significancia e Danos estruturais",
       x = "Danos estruturais", y = "Significancia")+
  theme_minimal()+
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 12, face = "bold"),
    axis.title.y = element_text(size = 12, face = "bold"),
    axis.text = element_text(size = 11)
  )
```



Produndidade vs. Danos estruturais

```
dados %>%  
  ggplot(aes(x = mmi, y = depth)) +  
  geom_point(alpha = 0.5, color = "blue") +  
  labs(title = "Relacao entre Profundidade e Danos estruturais",  
        y = "Profundidade (km)", x = "Danos estruturais")+  
  scale_x_continuous(limits = c(0.5, 9.5), breaks = 1:9) +  
  scale_y_reverse() +  
  theme_minimal()+  
  theme(  
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),  
    axis.title.x = element_text(size = 12, face = "bold"),  
    axis.title.y = element_text(size = 12, face = "bold"),  
    axis.text = element_text(size = 11)  
  )
```



Clustering geográfico simples (K-means)

```
set.seed(42)

proc <- dados %>%
  select(latitude, longitude, magnitude) %>%
  filter(!is.na(latitude) & !is.na(longitude) & !is.na(magnitude))

X <- proc %>% select(latitude, longitude, magnitude) %>% as.matrix()
X_scaled <- scale(X)

k <- 4
km <- kmeans(X_scaled, centers = k, nstart = 20)

proc$zone <- factor(km$cluster)

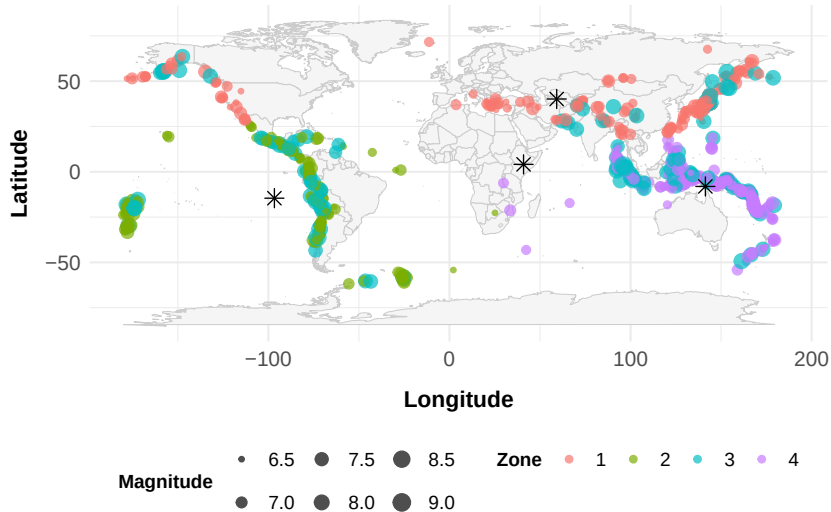
centroids <- proc %>%
  group_by(zone) %>%
  summarise(
    cen_lat = mean(latitude),
    cen_long = mean(longitude),
    cen_mag = mean(magnitude),
    n = n() )
```

Clustering geográfico simples (K-means)

```
world <- map_data("world")

ggplot() +
  geom_polygon(data = world, aes(x = long, y = lat, group = group), fill = "
    gray96", colour = "gray80", size = 0.2) +
  geom_point(data = proc, aes(x = longitude, y = latitude, color = zone, size =
    magnitude), alpha = 0.7) +
  geom_point(data = centroids, aes(x = cen_long, y = cen_lat), color = "black",
    shape = 8, size = 3) +
  scale_size_continuous(range = c(1, 4)) +
  coord_quickmap() +
  labs(title = paste0("K-means (k=", k, ") - Zonas geograficas"),
    x = "Longitude", y = "Latitude", color = "Zone", size = "Magnitude") +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    axis.title = element_text(size = 12, face = "bold"),
    axis.text = element_text(size = 11),
    legend.position = "bottom",
    legend.text = element_text(size = 10),
    legend.title = element_text(size = 10, face = "bold") )
```

K-means (k=4) – Zonas geográficas

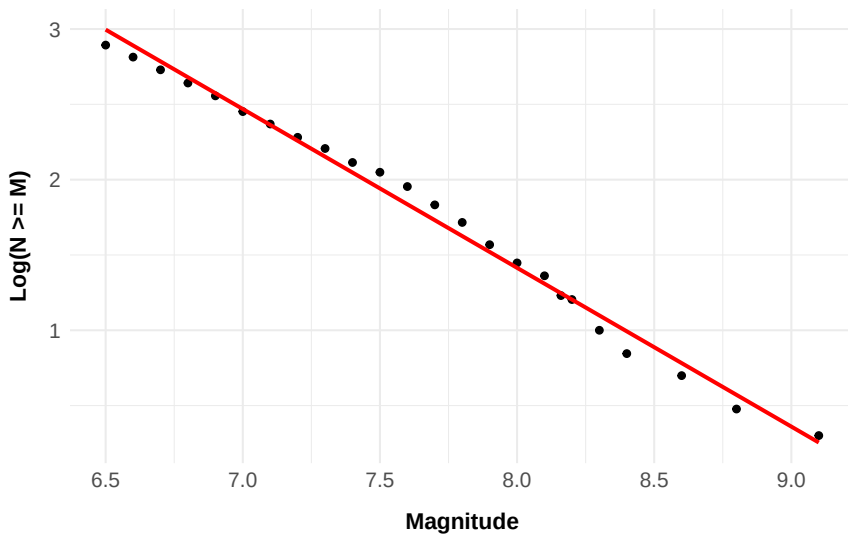


Gutenberg-Richter

```
gr_data <- dados %>%
  count(magnitude) %>%
  arrange(desc(magnitude)) %>%
  mutate(
    cumsum_n = cumsum(n),
    log_n = log10(cumsum_n)
  )

ggplot(gr_data, aes(x = magnitude, y = log_n)) +
  geom_point() +
  stat_smooth(method = "lm", formula = y ~ x, se = FALSE, color = "red") +
  labs(title = "Lei de Gutenberg-Richter", y = "Log(N >= M)", x = "Magnitude")+
  theme_minimal()+
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 12, face = "bold"),
    axis.title.y = element_text(size = 12, face = "bold"),
    axis.text = element_text(size = 11)
  )
```

Lei de Gutenberg–Richter

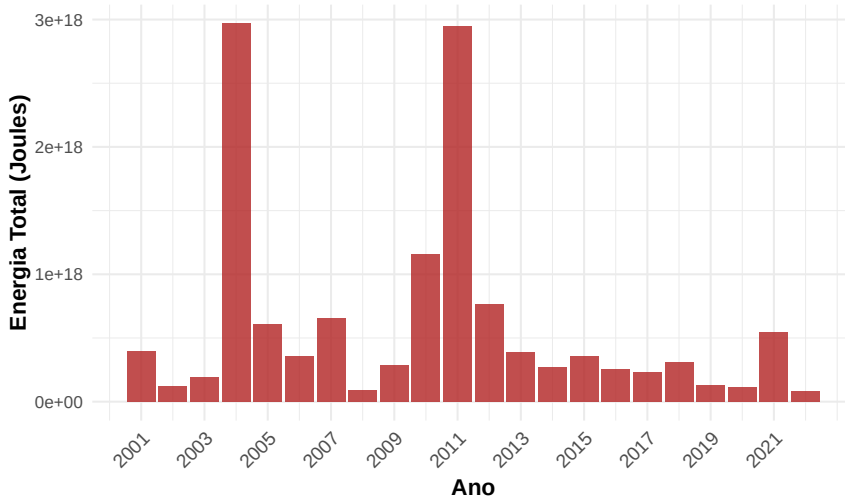


Energia liberada

```
# Formula de Gutenberg-Richter (Energia em Joules): log E = 4.8 + 1.5M
dados <- dados %>%
  mutate(energy = 10^(4.8 + 1.5 * magnitude))
dados %>%
  group_by(Year) %>%
  summarise(total_energy = sum(energy, na.rm = TRUE)) %>%
  ggplot(aes(x = Year, y = total_energy)) +
  geom_col(fill = "firebrick", alpha = 0.8) +
  scale_y_continuous(labels = scales::scientific) +
  labs(
    title = "Energia Sismica Total Liberada por Ano",
    subtitle = "A escala exponencial destaca anos com eventos extremos (ex:
      2004, 2011)",
    y = "Energia Total (Joules)", x = "Ano") +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 12, face = "bold"),
    axis.title.y = element_text(size = 12, face = "bold"),
    axis.text.x = element_text(size = 10, angle = 45, hjust = 1) )
```

Energia Sismica Total Liberada por Ano

A escala exponencial destaca anos com eventos extremos (ex: 2004, 2011)



Séries temporais

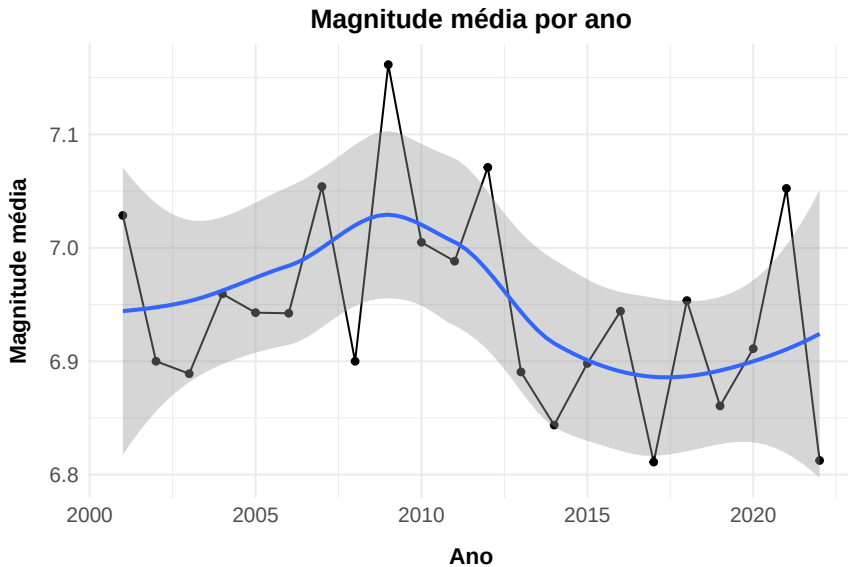
```
annual_summary <- dados %>%
  group_by(Year) %>%
  summarise(
    n_events = n(),
    mean_magnitude = mean(magnitude, na.rm = TRUE),
    sd_magnitude = sd(magnitude, na.rm = TRUE),
    mean_depth = mean(depth, na.rm = TRUE),
    sd_depth = sd(depth, na.rm = TRUE),
    n_tsunami = sum(as.numeric(as.character(tsunami)) == 1, na.rm = TRUE)
  ) %>%
  mutate(
    prop_tsunami = n_tsunami / n_events
  ) %>%
  arrange(Year)
```

Resumo anual (magnitude e profundidade)

Year	n_events	mean_magnitude	sd_magnitude	mean_depth	sd_depth	n_tsunami	prop_tsunami
2001	28	7.03	0.49	40.56	31.30	0	0
2002	25	6.90	0.43	37.46	48.10	0	0
2003	31	6.89	0.41	34.42	34.00	0	0
2004	32	6.96	0.53	41.98	57.38	0	0
2005	28	6.94	0.50	76.10	108.45	0	0
2006	26	6.94	0.52	30.22	29.44	0	0
2007	37	7.05	0.54	49.48	72.41	0	0
2008	25	6.90	0.34	35.23	29.47	0	0
2009	26	7.16	0.47	72.84	124.51	0	0
2010	41	7.00	0.45	49.52	97.26	0	0
2011	34	6.99	0.51	46.06	46.45	0	0
2012	31	7.07	0.53	69.81	111.80	0	0
2013	53	6.89	0.42	97.76	169.92	34	0.642
2014	48	6.84	0.40	75.39	133.28	40	0.833
2015	53	6.90	0.40	100.33	183.22	33	0.623
2016	43	6.94	0.42	93.22	151.50	31	0.721
2017	36	6.81	0.41	94.89	169.57	27	0.750
2018	43	6.95	0.42	145.81	223.69	33	0.767
2019	33	6.86	0.36	105.18	165.35	26	0.788
2020	27	6.91	0.40	81.24	150.76	15	0.556
2021	42	7.05	0.48	70.55	121.20	33	0.786
2022	40	6.81	0.28	128.43	211.16	32	0.800

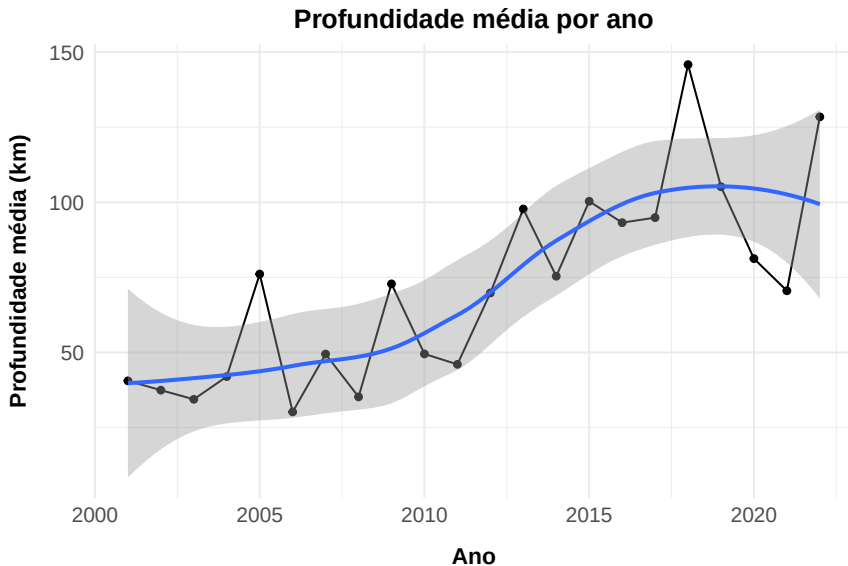
Magnitude média por ano

```
ggplot(annual_summary, aes(x = Year, y = mean_magnitude)) +  
  geom_line() + geom_point() +  
  geom_smooth(method = "loess", se = TRUE) +  
  labs(title = "Magnitude media por ano", y = "Magnitude media", x = "Ano") +  
  theme_minimal()+  
  theme(  
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),  
    axis.title.x = element_text(size = 12, face = "bold"),  
    axis.title.y = element_text(size = 12, face = "bold"),  
    axis.text = element_text(size = 11)  
  )
```



Profundidade média por ano

```
ggplot(annual_summary, aes(x = Year, y = mean_depth)) +  
  geom_line() + geom_point() +  
  geom_smooth(method = "loess", se = TRUE) +  
  labs(title = "Profundidade media por ano", y = "Profundidade media (km)", x =  
        "Ano") +  
  theme_minimal()+  
  theme(  
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),  
    axis.title.x = element_text(size = 12, face = "bold"),  
    axis.title.y = element_text(size = 12, face = "bold"),  
    axis.text = element_text(size = 11)  
  )
```



Conclusão

- Maioria dos eventos registrados no período apresentaram magnitude menor de 7.0;
- Aumento da profundidade média ao longo dos anos
- Padrões Geográficos Claros: A clusterização (K-means) confirmou que o risco se concentra nas falhas tectônicas conhecidas (Círculo de Fogo), validando a qualidade dos dados.
- A análise de energia demonstrou que poucos eventos extremos (como 2004 e 2011) são responsáveis pela maior parte da liberação de energia sísmica do século.
- Relação Dano x Intensidade: Confirmamos a forte correlação entre impacto estrutural (MMI) e a magnitude, servindo como bons preditores de risco humanitário.